

HDL style

Synthesizability lint asks whether your code can become gates

Naming and block keywords

- Clock nets should be named `clk` or end with underscore `clk` so every reader spots the timing
- Active-low reset should use `rst_n` or `reset_n`
- For edge-triggered registers, SystemVerilog prefers `always_ff` over plain `always` at `posedge`
- For combinational procedural logic, prefer `always_comb` over `always` at `star`
- For new RTL, prefer `logic` over `reg` when your toolchain supports it
- These are info-level hints in the teaching linter

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Naming / style checker

Teaching style hints from the HDL engine (lintStyle): `clk` / `rst_n` naming, `always_ff` / `always_comb`, and `logic` over `reg`. The same API powers live Problems assist in the simulator IDE. Starter: clean FF with `clk + rst_n`.

Starter example: clean `always_ff` with `clk + rst_n`. Try snippets that emit style hints, then fix them.

Challenge 0 / 22 cleared

Quiz: what is this: This lab's lintStyle API is...

☐ a teaching / assist style pack — not Verible or a company style guide

☐ identical to Vivado lint

☐ only a formatter

☐ a synthesis tool

Show hint

Check

Next

Load challenge setup

Idle

Quiz: what is this

Quiz: clock name

Quiz: reset name

Quiz: always_ff

Quiz: always_comb

Quiz: logic vs reg

Quiz: IDE assist

Lint: clean FF

Lint: name-clk

Lint: name-rst

Lint: prefer-always-ff

Lint: prefer-always-comb

HDL style lab starter

Real Verilog practice

```
wsl - module17-hdl-style/examples/navigation

# Track A - real Unix
# real Verilog session (Track A)

$ cat examples/style_demo/style_demo.v
// style_demo.v - conventional naming and block style
module ff_styled(
    input      clk,
    input      rst_n,
    input      d,
    output reg  q
);
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) q ≤ 1'b0;
        else       q ≤ d;
    end
endmodule

module and2_styled(
    input a,
    input b,
    output y
);
    assign y = a & b;
endmodule

// clock_bad - style warnings (commented)
// module clock_bad(input clock, d, output reg q);
//     always @(posedge clock) q ≤ d;
// endmodule

$ iverilog -t null examples/style_demo/style_demo.v
(syntax check passed)
```

Real Verilog - styled flip-flop and combo

Pitfalls to watch

- Do not confuse style hints with synthesizability errors
- Do not rename signals randomly; pick one convention per project and stick to it
- The keyword choice documents intent
- It teaches the categories you will see in real review tools

Your turn

- Complete the checklist for at least one track, ideally both
- In the browser lab, load the clock-not-clk snippet and fix it until name-clk disappears
- In real Verilog
- When you finish